Because the cranium is a closed compartment, an increase in volume in any of the structures or components located within it will result in an increase in intracranial pressure (ICP). The first effect of an increase in the volume of a component will be a decrease in cerebral blood flow.

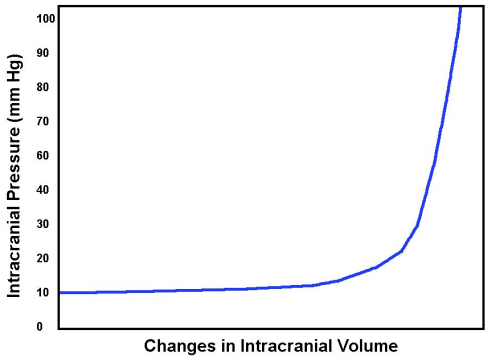
The cerebral perfusion pressure (CPP) equals the mean arterial pressure (MAP) minus the ICP (or CVP, whichever is higher). If the ICP increases faster than the MAP, the perfusion to the brain is compromised and the brain can become ischemic. The normal ICP is less than 10 mmHg. The definition of intracranial hypertension, however, can be consistently found as a sustained pressure above 15 mmHg.

If an increase in ICP is substantial enough, the brain can herniate through the cingulate gyrus, the uncinate gyrus, at the level of the cerebellar tonsils through the foramen magnum, or through an opening in the skull, leading to rapid deterioration of the patient's neurologic status.

There are three major causes of cerebral edema: vasogenic, cytotoxic, and interstitial. Vasogenic edema is the result of a breakdown in the blood-brain barrier which allows protein to move into the extracellular tissue of the brain (thus drawing in water by increased oncotic pressure). Cytotoxic edema is the result of neuronal damage from hypoxia or ischemia, which increases the amount of sodium and water in the brain cells, leading to increased brain cell volume. Interstitial edema can result from the entry of cerebrospinal fluid into the interstitium due to hydrocephalus. Acute decreases in serum osmolality from water intoxication can also produce cerebral edema.



Cerebral perfusion pressure = MAP – ICP



The intracranial contents possess a normal elastance, therefore patients lacking intracranial disease are not affected by transient increases in ICP (those seen with Valsalva maneuvers or coughing). The normal ICP is maintained. Radical increases in ICP are also not seen with minute increases in the intracranial volume. There is a limit, however, to this normal elastance, once the intracranial contents have reached capacity. Dramatic increases in ICP are seen when the intracranial volume has reached capacity, as seen with an enlarging mass, tumor, or blood. Intracranial volume remains relatively constant during beginning stages of tumor growth or hemorrhage.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 594.

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